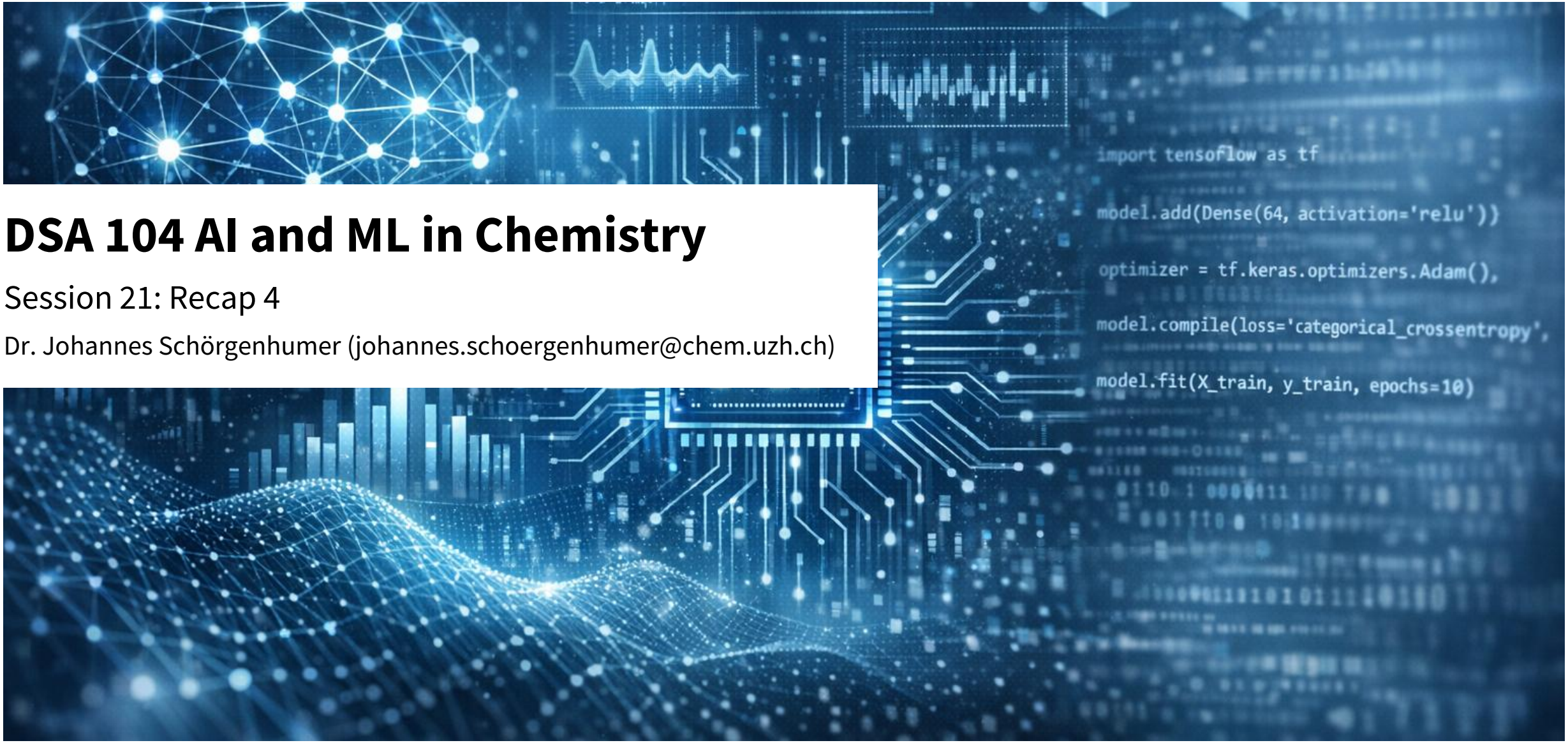
Universität Zürich[UZH]

Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 21: Recap 4

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Dr. Stephan Guttinger, University of Exeter (UK):

The Ethics of Automated Research

DSA 104 – Virtual guest lecture

What are some core ethical issues with AI and Automation?

What does research ethics look like when experiments are planned, performed and analysed by multi-agent AI systems instead of humans?

What is the role of the human in automated research?

Join Dr. Stephan Guttinger's guest lecture to find out!

When?

May 13, 13:00 – 14:45

Where?

Hybrid lecture: **Y19-J-94**

MS Teams: Scan the QR Code

Points for today

Light content: R2 as metric

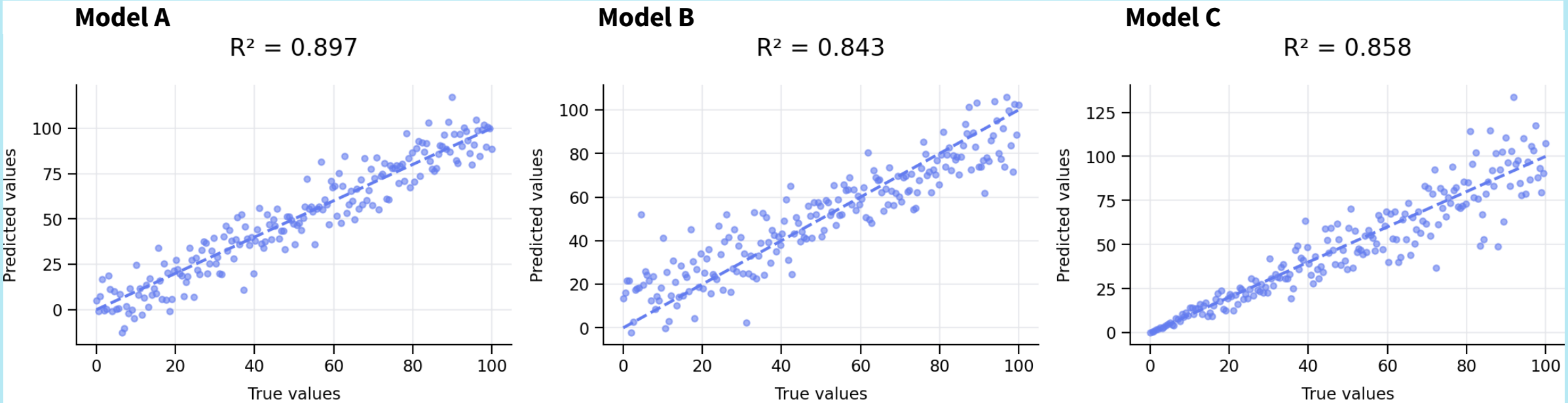
Quiz time!

Light content 2: Bayesian optimisation

Discussion

Model evaluation: metrics

Discuss: How does the R2 capture the errors in the following models?



Metrics: R2

Measures how much **variance** in the target is explained by the model!

— *"How much better is the model than predicting the mean?"*

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y}_i)^2}$$

Similar as MSE definition, but correlated to the squared deviation from the mean (variance!)

Result: Higher R2 does not mean that the model is more accurate!!

E.g. 1 error of +100 and 9 prefect values will give the same R2 as 10 errors with the magnitude 31.6!

Problems with the R2 as metric

- **Hides error magnitude:**
  - Does not tell you how wrong the model is!
  - No comparability between different models / targets (different magnitudes of errors…)
- **Insensitive to biases:**
  - Model can systematically over- and underpredict, despite high R2
    - Always predicts 10% too high
    - Still explains variance well, i.e. still high R2
- **Depends on target variance:**
  - Even good models get low R2 if the target has high variance
  - Weak models can look good
- **Encourages overfitting:**
  - R2 usually improves if you add more features on training data!

Model	Error structure
A	Small random noise everywhere
B	Large errors only at high values
C	Strong systematic bias
D	Occasional catastrophic failures

How and when to use the R2 as metric:

When is R2 useful?

— You care about **overall variance explanation**

— Errors are **approximately symmetric**

— You’re doing **exploratory or scientific** modeling

Best practice:

— Use only to compare on the same data

— Combine with other error metrics

Quiz time!

Definitions:

— Attention function

— Forward pass

— Convolution layer

— Zero Padding

— Pooling

— Supervised ML – Unsupervised ML – Reinforcement Learning

— Inductive Transfer learning

— Catastrophic forgetting

— Zero-shot learning

— SDL

Quiz time – two truths and a lie!

CNNs

— A. Convolutions exploit spatial locality by using shared weights across an input feature map.

— B. CNNs are inherently translation equivariant but not translation invariant unless pooling or global aggregation is used.

— C. CNNs require fixed-size inputs because convolution operations cannot handle variable input dimensions.

Transfer learning

— A. Transfer learning always improves performance compared to training from scratch.

— B. Transfer learning is most effective when the source and target domains share similar low-level features.

— C. Freezing early layers and fine-tuning later layers is a common strategy in vision transfer learning.

Foundation models

— A. Foundation models are typically pre-trained on large, diverse datasets and adapted to downstream tasks.

— B. Foundation models usually require task-specific architectures to generalize across domains.

— C. Prompting is an alternative to fine-tuning for adapting foundation models to new tasks.

Quiz time – two truths and a lie!

Time Series Modelling

— A. Temporal autocorrelation violates the i.i.d. (independent identically distributed) assumption common in many ML models.

— B. Shuffling time-series data before train-test split can lead to overly optimistic performance estimates.

— C. Stationarity means that the time series does not change its values over time.

Reinforcement learning

— A. In reinforcement learning, the reward signal is often delayed rather than immediately informative.

— B. Model-based RL explicitly learns environment dynamics and can improve sample efficiency.

— C. Exploration is unnecessary once an optimal policy has been found.

Generative models

— A. Variational Autoencoders (VAEs) optimize a lower bound on the data log-likelihood.

— B. Generative Adversarial Networks (GANs) model the data distribution by specifying and optimizing an explicit likelihood function.

— C. Diffusion models generate samples by reversing a gradual noising process.

Quiz time – two truths and a lie!

Applicability Domains & Model Validity

— A. Model predictions are most reliable when inputs lie within the training data distribution.

— B. Applicability domains can be estimated using distance-based or uncertainty-based methods.

— C. High validation accuracy guarantees reliable predictions outside the training domain.

DL and Timeseries

— A. Standard cross-validation without temporal ordering is best practice for time series.

— B. Temporal convolutional networks can model long-range dependencies using dilation.

— C. Recurrent neural networks explicitly model sequential dependencies.

Applicability and RL

— A. RL is sensitive to distribution shift between logged data and learned policies.

— B. Safety constraints are often critical when deploying RL systems in real-world environments.

— C. RL agents generalize well outside their training environments by default.

Bayesian optimisation (BO)

Many **problems in** chemistry **and life sciences** share three key properties:

— Experiments are **expensive** (wet-lab assays, synthesis, simulations)

— Objective is a **black box** (no analytical gradients)

— Data is **scarce** (tens, not millions, of experiments)

Problem: **Function is not available in closed form**, i.e. expensive to solve, e.g.

— Result of a simulation

— Results of hyperparameter optimisation of an ML model

— Experimental results (Design of Experiments!)

BO targets exactly that regime!

Examples from the life sciences:

— Maximising reaction yield or selectivity

— Optimising protein expression conditions

— Tuning drug formulation parameters

— Hyperparameters of costly molecular simulations

BO: problem setting

We want to solve:

$$x^* = \arg\max_x f(x)$$

Where:

— x are the decision variables, $x \in \mathbb{R}^d$ (e.g. reaction conditions)

— x^* is the optimal input

— $f(x)$ is unknown, noisy or expensive to evaluate

Basically, **maximisation problem**, but each evaluation of $f(x)$ might take hours/days/weeks (e.g. conduct an experiment)

Solution: Approximate iteratively!

H. J. Kushner, *J. Basic Eng*. **1964**, *86*, 97-106, https://doi.org/10.1115/1.3653121
J. Mockus, “On a Bayesian method for seeking an extremum”, *Automatika i vychislitelnaja tekhnika, N3,* **1972.**

BO: Core idea

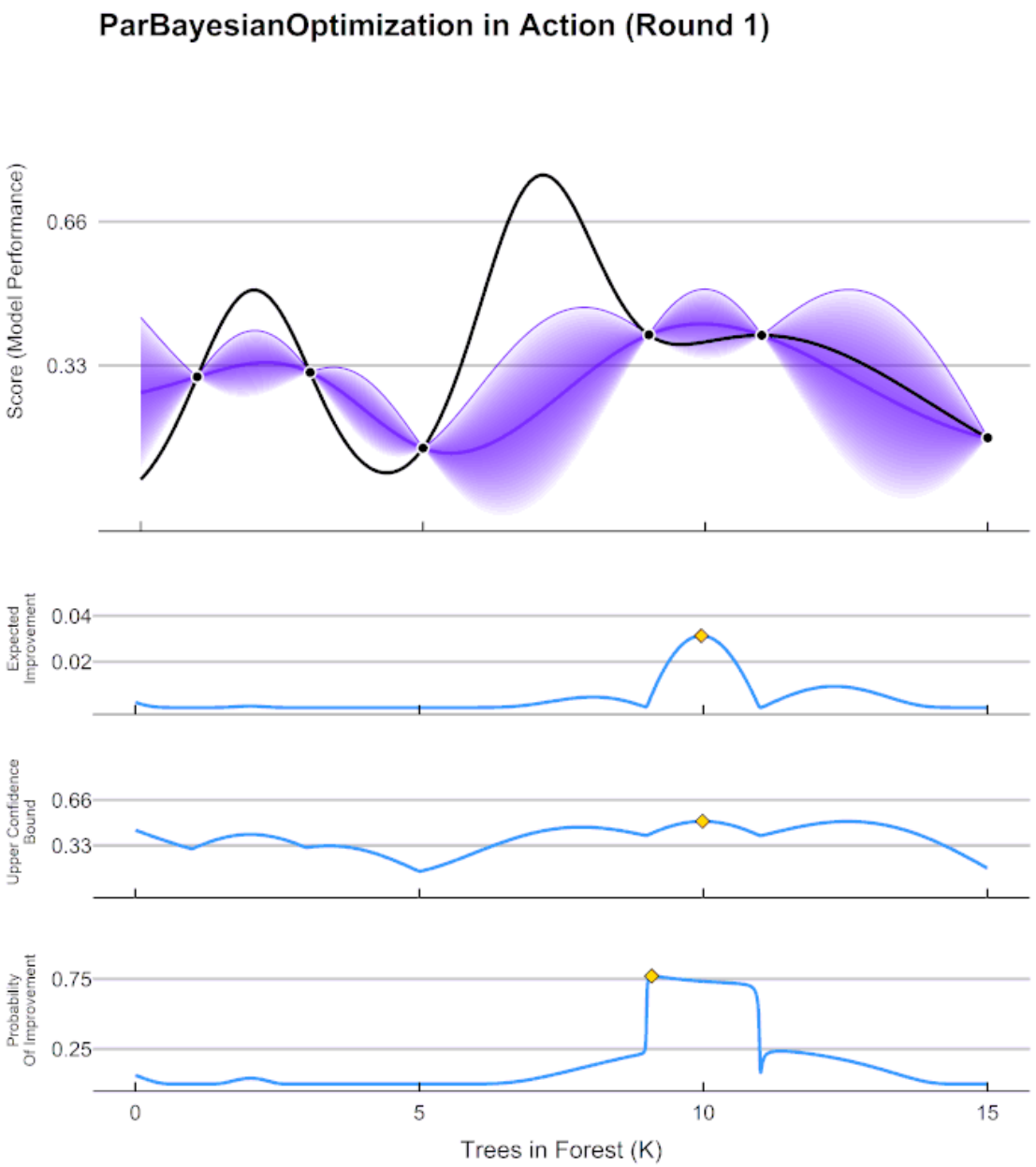
Bayesian optimisation **alternates between**:

- **Surrogate model**
  - Probabilistic model of f(x) (usually a Gaussian process)

- **Acquisition function**
  - Decides where to sample next, balancing:
    - Exploration (high uncertainty)
    - Exploitation (high predicted value)

Loop:

Fit surrogate → Optimise acquisition → Run experiment → Update model

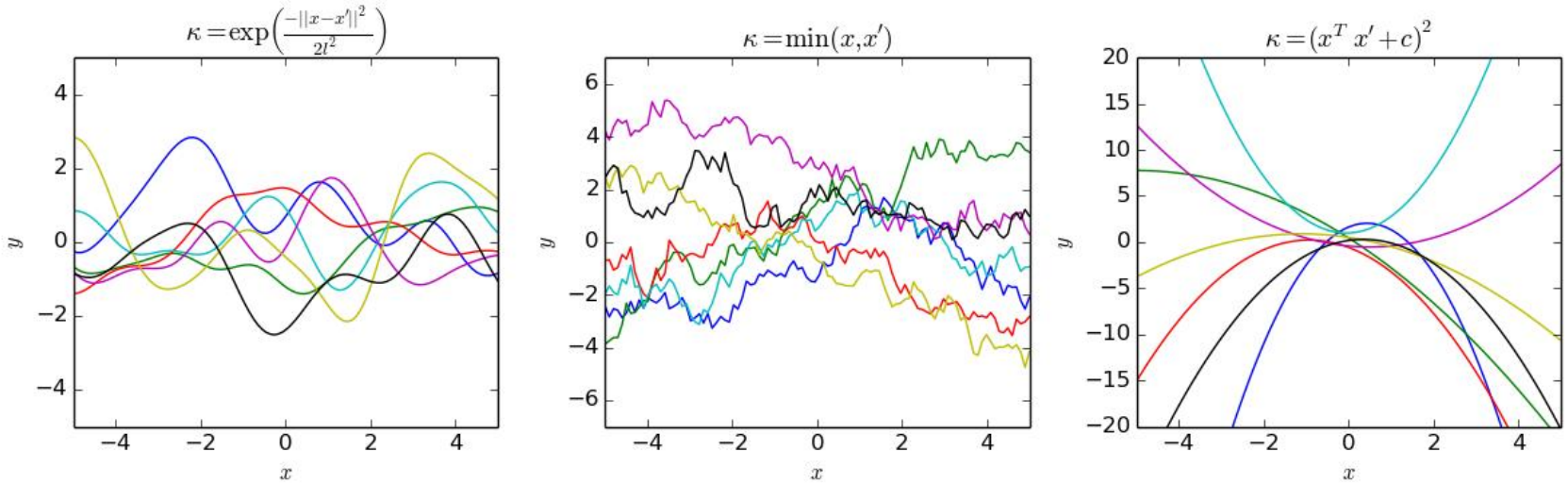


Animation by AnotherSamWilson – Own work, CC BY-SA 4.0,
https://commons.wikimedia.org/w/index.php?curid=84842869

BO: Gaussian Processes

A Gaussian Process (GP) is a distribution over functions.

- For any point x, the GP predicts:
  - **Mean**: expected objective value
  - **Variance**: uncertainty



- Key properties:
  - Works well with **small datasets**
  - Naturally quantifies **uncertainty**
  - Flexible via **kernel** choice (e.g. RBF is common)

- E.g. in chemistry:
  - Smooth changes in yield vs temperature
  - Correlated effects of solvents or concentrations

Different kernels functions: Image by Cdipaolo96 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=47589358

BO Decision Rules: Acquisition functions

Popular choices:

- **Expected Improvement** (EI):
  - "How much improvement over the best observed value do we expect?"
  - Balances exploration and exploitation naturally
  - Especially popular in **experimental sciences**

Improvement: $I(x) = \mu(x) - y_{best} - \xi$

…with ξ as parameter for exploration ("beat by margin ξ")

Standardise as: $Z = \frac{I(x)}{\sigma(x)}$

$$EI(x) = \underbrace{(\mu - ybest - \xi)\Phi(Z)}_{\text{Exploitation (likely improvement)}} + \underbrace{\sigma\phi(Z)}_{\text{Exploration (uncertainty bonus)}}$$

- **Upper Confidence Bound** (UCB):
  - "Act as if the function is as good as plausibly possible."
  - High mean: "looks good"
  - High uncertainty: "could be good"

$$UCB(x) = \mu(x) + \kappa\sigma(x)$$

…with κ as exploration parameter

BO Exercise

Have a look at the *BO.ipynb* notebook.

Run the code and discuss the points in the end. Test how noise and exploration parametrisation influence the model.

Discussion time

Which parts of the contents so far do you find most important?

Do you already have any applications in mind where you would like to try out some things?

Think of the bigger picture:

— AI in our everday world

— AI in the media / politics

— How is AI reshaping our society?

— AI as matter of security (both virtual an real world)

Discuss risks and advantages!